A Lightweight Robotic Ultrasound System With Hybrid Force Control and Modular Force-Range Switching

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***Abstract*—Ultrasound imaging provides valuable, radiation-free soft tissue visualization, with its portability enabling diagnostics in diverse settings. However, image quality is operator-dependent, where variations in probe contact force significantly compromise diagnostic consistency. To address this issue, we present a novel general-purpose robotic ultrasound system. The system features a lightweight end-effector, hybrid force control, and a modular mechanism for selectable force ranges. A tunable constant-torque mechanism, designed via stiffness matching, dampens sudden tremors. This torque is translated into a precise, constant contact-force through modular winding reels. A closed-loop control system with comprehensive compensation for gravity, hysteresis, and friction enables real-time deformation tracking. Experimental studies on silicone phantoms and human volunteers demonstrate the system's ability to acquire high-quality scans with exceptional repeatability and operational ease. During abdominal ultrasound scanning with deep breathing, the proposed system successfully suppresses oscillations to a magnitude of 0.10 mm. The reported robotic ultrasound system offers an effective solution for consistent and comfortable ultrasound examinations.**

***Index Terms*—Compliant mechanism, force control, medical robot, modular design, ultrasound (US) imaging.**

I. Introduction

ULTRASOUND (US) imaging has become an essential modality in modern medical diagnostics, owing to its unique advantages of being radiation-free, real-time, low-cost, and highly convenient [1], [2], [3]. It plays a critical role in diagnosing conditions of the thyroid, breasts, abdominal organs, and in gynecology and obstetrics. These merits also make it particularly desirable for bedside and outdoor diagnostics, benefiting emergency treatments [4]. However, the image quality in conventional manual US scanning is highly dependent on operator skill. This dependence leads to substantial discrepancies in image quality and diagnostic conclusions across different operators, potentially resulting in misdiagnosis and imposing high operator training costs [5], [6]. This issue is more pronounced in rural and remote areas [7].

Therefore, robotic assistance is highly desirable to facilitate US scanning procedures. To meet the demands of autonomous robotic US acquisitions [8], remote examination support [9], and standardized US examinations [10], robotic ultrasound systems (RUSSs) face two primary challenges: anatomical alignment and contact force control. Accurate anatomical alignment is indispensable for the development of fully autonomous RUSSs [11]. This work concentrates on the other core challenge: contact force control between the US probe and biological tissue, as contact force exerts substantial impact on the US imaging quality [12]. An insufficient force leads to poor acoustic coupling and blurred US images, whereas excessive force causes severe tissue deformation and carries a risk of misdiagnosis. Unstable contact force leads to persistent image variations (positions and deformations of measured tissues [13]), which further complicates clinical diagnostic procedures. Furthermore, sustaining a precise and consistent contact force during the entire scanning process remains a formidable challenge. In addition, contact force requirements differ substantially across various scanning targets, typically ranging from 2.4 to 12 N, and can even reach triple this value in certain specialized applications [14]. Accordingly, a high-performance RUSS must be capable of reliable operation over this broad force range. Based on the literature-reported force variation of 11.6 $\pm$ 2.47 N during manual fetal US scans conducted by sonographers [15], a force variation within $\pm$20% of the nominal contact force magnitude is defined as the maximum acceptable threshold for the proposed system.

To address the challenge of contact force control, numerous force control strategies have been reported in the literature [16], [17], [18], [19], [20], which are broadly classified into active and

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passive approaches. Active control achieves force regulation by dynamically adjusting actuator positions in response to sensor feedback. For example, Gilbertson [21] and Sai [22] employed a proportional-derivative controller and an adaptive proportional–integral–derivative controller, respectively, to attenuate contact force variations during US scanning. Wang et al. [23] introduced an admittance control framework for regulating six degree-of-freedom forces using a robotic arm equipped with a 6-D wrist force sensor. Fu et al. [24] proposed a variable impedance controller with a shared control strategy to assist force control in remote US scanning. Furthermore, interposing a spring between the actuator and force sensor can attenuate the rate of force change, thereby enhancing the control performance of active control systems [25], [26], [27]. In contrast, passive control relies on predefined elastic elements, such as springs or compliant mechanisms, to provide constant force. For example, Wang et al. [28] developed a force adjustment mechanism for a robotic arm based on the characteristics of a torsion spring. Kuo et al. [29] designed a constant-force end-effector using a pneumatic actuator, which adjusted the force magnitude from 4.10 to 7.53 N. Their design used a ratchet structure to apply various preloads to a torsion spring, enabling a tunable constant-force on the US probe. Tsumura et al. [30], [31] developed robotic fetal US scanning platforms that maintained contact force via a constant-force spring. In addition, existing literature suggests that constant-force compliant mechanisms are well-suited for passively controlling contact forces [32], [33], [34].

However, both approaches suffer from inherent limitations. Owing to their dependence on closed-loop feedback, active control systems are prone to control delays caused by limited bandwidth [35], [36]. This makes it difficult to suppress force overshoots during rapid movements, thereby introducing potential safety hazards. Furthermore, the demand for high-bandwidth sensors and actuators elevates the overall system cost. By contrast, passive control exhibits distinct merits, including fast response, low cost, and inherent immunity to force overshoot. Nevertheless, the adaptability of passive systems is restricted by fixed mechanical components, which limits its capacity to meet patient-specific demands regarding force magnitude and scanning angle. To combine the advantages of both control schemes, Wu et al. [37] developed a hybrid force control strategy. In this design, active control regulates the force magnitude and deformation of a passive mechanism (PM) to compensate for low-frequency vibrations (e.g., physiological breathing motions), whereas passive control damps high-frequency transient vibrations (e.g., sudden tremors). This strategy effectively stabilizes the contact force throughout US scanning, thus enhancing both image quality and scan repeatability. Nevertheless, this particular RUSS is bulky and heavy, requiring mounting on a robotic arm and compromising the portability of the US imaging system. Moreover, its constant-force range is adjusted by physically replacing PMs with varying thicknesses. This entails fabricating multiple RUSS prototypes or conducting complicated disassembly-reassembly processes to satisfy diverse scanning requirements, which is both financially costly and labor-intensive. Accordingly, there is an urgent demand for a general-purpose RUSS integrated with a lightweight end-effector and a modular, switchable force ranges enabled by hybrid force control.

To this end, this article presents the design, development, and experimental validation of a novel RUSS. The proposed system synergistically integrates a lightweight end-effector, hybrid force control architecture, and modular range switching mechanism, thereby enhancing adaptability, image quality, and operational efficiency for diversified US scanning scenarios. The main contributions of this work are summarized as follows.

1) *A novel RUSS with a lightweight end-effector:* The developed RUSS adopts a compact and lightweight structural design, enhancing its universality for both robotic arm integration and direct handheld operation. This superior portability enables bedside and field diagnostic scenarios, and the nonintrusive form factor effectively improves patient compliance during clinical examinations.
2) *A hybrid force control strategy for robust performance:* The system integrates a passive constant-torque mechanism for tremor absorption with an active control module realizing full compensation for gravity, hysteresis, and friction. This hybrid approach sustains stable probe-tissue contact force against diverse external disturbances, thereby markedly enhancing the quality and repeatability of US imaging.
3) *A modular system for rapid force range reconfiguration:* A novel modular winding reel mechanism is proposed to achieve rapid switching of the system constant-force range, satisfying the diversified force demands across various US scanning tasks. This modular architecture improves operational efficiency by shortening setup time and lowering overall system costs for multifarious scanning applications.

Specifically, the PM serving as the core component of the RUSS was designed via a stiffness matching strategy [38], [39], enabling variable constant-torque output regulated by the angular displacement between its internal rotors. Such torque is further converted into adjustable constant-force outputs through interchangeable modular winding reels with various dimensions. A closed-loop control framework is deployed to accurately regulate the angular displacements of the PM. The proposed control scheme integrates dedicated compensators to suppress the adverse impacts of gravity, hysteresis, and friction, and utilizes a proportional-integral controller (PIC) to improve the real-time tracking performance of the servo motors. Furthermore, the structural parameters of the PM are optimized via finite element analysis (FEA) and experimental calibration, followed by the fabrication of a fully functional prototype. Finally, the feasibility and overall performance of the proposed RUSS are systematically verified via a series of dedicated experimental tests.

The rest of this article is organized as follows. Section II elaborates on the working principle and mechanical analysis of the proposed RUSS. Section III introduces the parameter determination and optimization of the PM via FEA. Section IV presents the prototype fabrication and provides a thorough discussion on the experimental results. Finally, Section V concludes this article with concise remarks.

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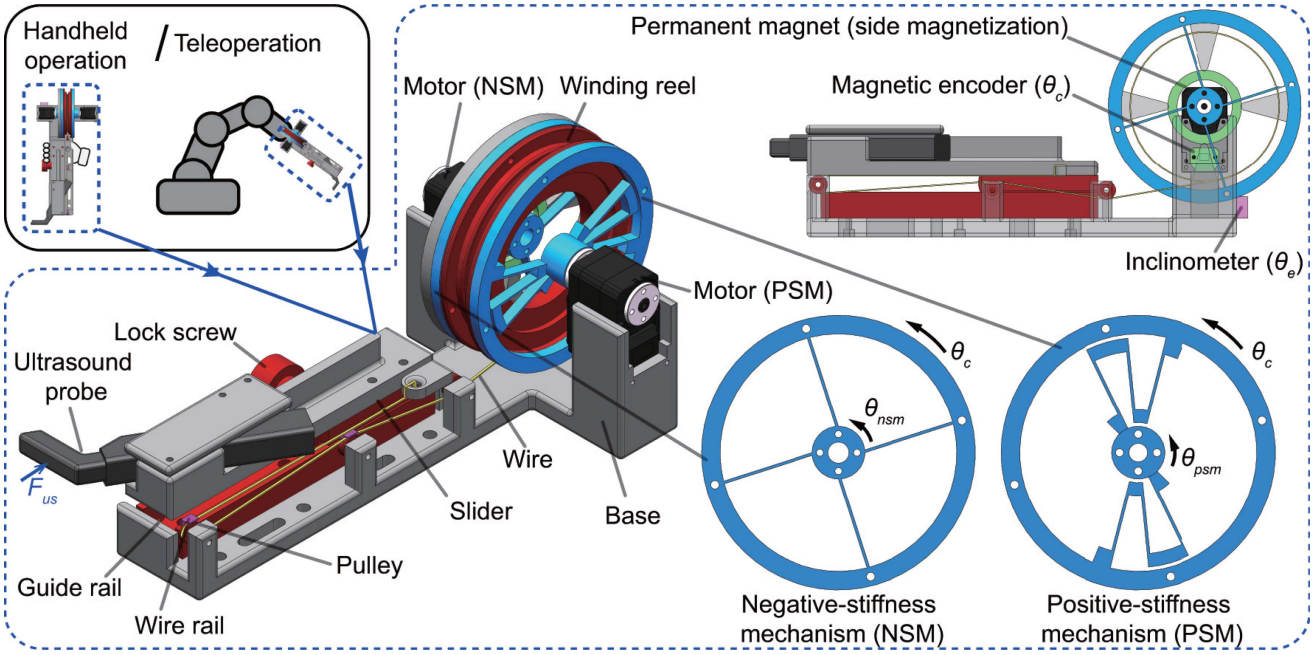


Fig. 1. Schematic diagram of the proposed RUSS supporting both handheld and teleoperated operations. The end-effector mainly consists of an NSM, a PSM, a winding reel, two servo motors, a magnetic encoder, a permanent magnet, a wire, a base, three pulleys, three wire rail, a slider with lock screw and its guide rail, an US probe, and several fasteners.

II. Mechanism and System Design

This section details the working principle and analytical foundation of the proposed RUSS.

A. Overall Mechanism Design

The computer-aided design (CAD) model of the proposed RUSS is shown in Fig. 1. The system's core is a constant-torque mechanism that combines a negative-stiffness mechanism (NSM) and a positive-stiffness mechanism (PSM) to create a constant-torque region, which effectively buffers sudden, short-duration disturbances. In comparison with the existing constant-force mechanisms [37], [38], [39], the proposed constant-torque mechanism demonstrates greater compatibility with conventional motors, as it does not need the incorporation of an additional linear slide module with the motor. This design significantly reduce the overall weight of the end-effector.

To maintain stable system operation within the predefined constant-torque regime despite large-amplitude vibrations from the scanned tissue surface, the mechanism utilizes two servo motors to actively regulate the angular deformations of the NSM (θ_n) and PSM (θ_p). These angles are derived as $\theta_n = \theta_{\text{nsm}} - \theta_c$ and $\theta_p = \theta_{\text{psm}} - \theta_c$, respectively, where θ_{nsm}, θ_{psm}, and θ_c are the angles measured by the encoders on the NSM, PSM, and winding reel. The relative angle between the mechanisms is defined as $\theta_{\text{pn}} = \theta_n - \theta_p$. The reaction torque τ_c from the mechanism is transmitted to the winding reel. Via a wire-driven transmission system guided by a slider and pulleys, the torque generated by the winding reel is converted into the axial contact force F_{us} applied at the US probe. Furthermore, during target tissue localization, the US probe may deviate from the normal direction of the scanning surface, particularly under handheld operation. Any resulting off-axis forces or torques will be constrained by the guide rail, as illustrated in Fig. 1. Thus, the axial acoustic coupling remains unaffected by such lateral probe motions.

The magnitude of the constant contact force F_c can be regulated by tuning θ_{pn}, which shifts the zero position of the PSM, as depicted in Fig. 3. This same principle adopted realize compensation for gravity, hysteresis, and friction via the modulation of θ_{pn}. Nevertheless, the feasible adjustment range of θ_{pn} is limited by the yield strength of the constituent materials. To surmount this constraint, the contact-force range of the system is reconfigured by replacing the modular winding reel with units of distinct radii R_w. In practice, conventional constant-force mechanisms necessitate complete redesign and reassembly to achieve considerable force magnitude variations. Accordingly, straightforward replacement of the modular winding reel represents a superior strategy, which improves cost-effectiveness and simplifies operational procedures.

The overall system design is categorized into the following three functional parts.

1) The passive constant-torque control part, which provides the fundamental force-regulation capability.
2) The modular torque-to-force conversion part, which translates torque into a linear contact force and enables range switching.
3) The active deformation control part, which uses the servo motors to maintain optimal system state and provide compensations.

The following analytical models describe the working principles of the proposed system.

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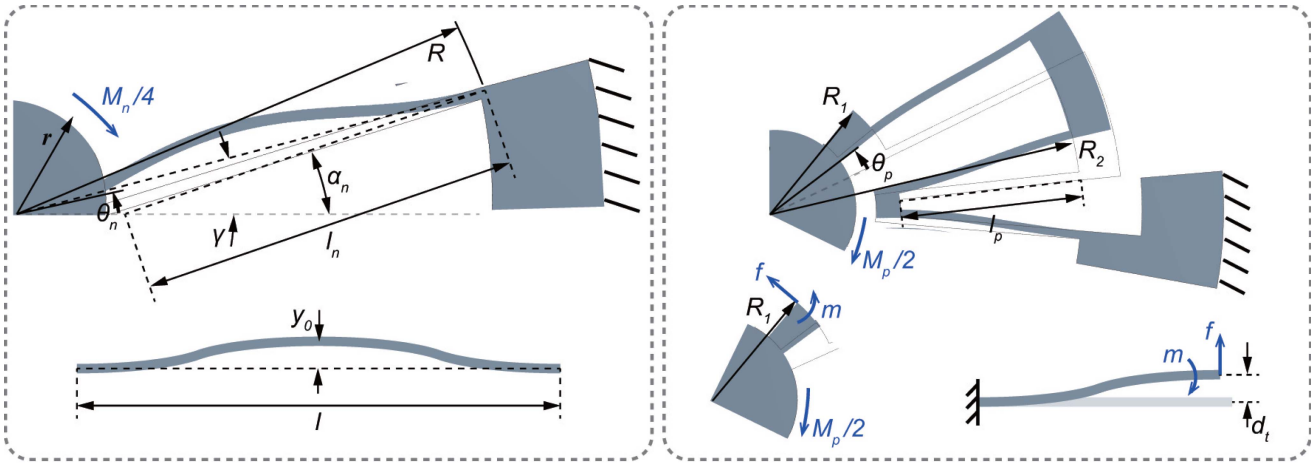


Fig. 2. Analyses of the (a) quarter-model of the NSM and (b) half-model of the PSM, demonstrating the deformation behaviors of submechanisms subjected to external torques.

B. Modeling of Passive Constant-Torque Control Part

The passive constant-torque control part, i.e., PM, integrates an NSM and a PSM to create a deformation region with a constant reaction torque. Both the NSM and PSM comprise an outer ring connected to a central cylinder by several specially designed beams.

Fig. 2(a) shows a quarter-model of the NSM. In this mechanism, the connecting beams are oriented at an angle α_n to the radial direction. When an angular deformation θ_n is applied, as depicted in Fig. 2(a), the beams undergo compressive buckling. This buckling effect generates a negative stiffness within a specific range of deformation. The resulting buckling strain energy U_n of an NSM beam can be expressed as

$$U_n = \frac{1}{2} E I_n \int_0^l (y_0 \sin''(\frac{\pi x}{l}))^2 dx \tag{1}$$

where E is the Young's modulus. I_n is the area moment of inertia of the buckling beam, which is calculated by $b_n h_n^3/12$. b_n and h_n are the out-of-plane thickness and in-plane width of the buckling beam, respectively. $y_0 \sin''(\frac{\pi x}{l})$ is the second derivatives of the first modal function of the buckling beam. l is the distance between two ends of the buckling beam. y_0 is the bulking displacement in the central point of the buckling beam.

The original in-plane length of the buckling beam (l_n) is related to y_0 via the arc length integral, which is expressed as

$$l_n = \sqrt{1 + y_0^2} E(i, \pi) \tag{2}$$

where $E(i, \pi)$ denotes the incomplete elliptic integral of the second kind and i represents the modulus ranging within the interval [0, 1].

Then, the length l can be derived from geometric considerations, as illustrated in Fig. 2(a)

$$l = \sqrt{(R\cos(\gamma - \theta_n) - r)^2 + (R\sin(\gamma - \theta_n))^2} \tag{3}$$

where R and r are the radii of the ring and central cylinder, respectively. The angle γ is determined by

$$\gamma = \arctan(\frac{l_n \sin\alpha_n}{l_n \cos\alpha_n + r}). \tag{4}$$

In addition, R can also be derived by the geometry relationship, which is expressed as

$$R = \sqrt{(l_n \cos\alpha_n + r)^2 + (l_n \sin\alpha_n)^2}. \tag{5}$$

Then, the reaction torque of the NSM (M_n) with a angular deformation can be calculated by

$$M_n = 4\frac{dU_n}{d\theta_n}. \tag{6}$$

Fig. 2(b) illustrates a half-model of the PSM. The mechanism features three beams aligned radially, each with an in-plane length l_p, connected in series. Consequently, all three beams experience the same translational deformation d_t. According to the boundary conditions, the following relationships are obtained:

$$\frac{f l_p^3}{3EI_p} - \frac{m l_p^2}{2EI_p} = d_t \tag{7}$$

$$\frac{f l_p^2}{2EI_p} - \frac{m l_p^2}{EI_p} = 0 \tag{8}$$

where f and m denote the force and the moment acting on the free end of the beam, as shown in Fig. 2(b). I_p is the area moment of inertia of the beam of PSM. which is determined as $b_p h_p^3/12$. b_p and h_p denote the out-of-plane thickness and in-plane width of the PSM beam, respectively.

Since θ_p corresponds to the sum of equivalent deflections of the three beams toward the center, as illustrated in Fig. 2(b), the relationship between θ_p and d_t is derived as follows:

$$\theta_p = d_t \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_1}\right) = \frac{d_t(2R_2 + R_1)}{R_1 R_2} \tag{9}$$

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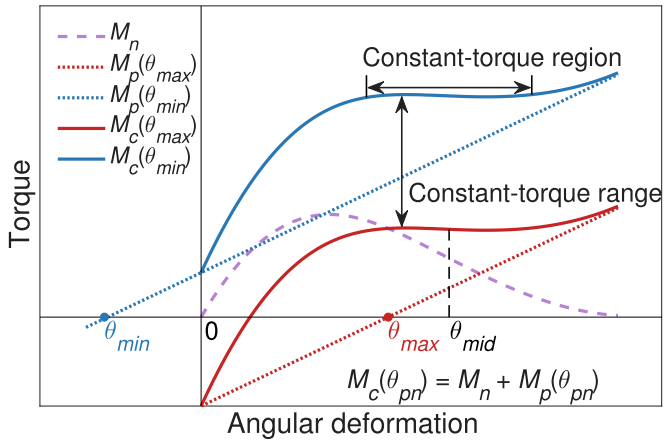


Fig. 3. Torque–deformation relations of the PM and its submechanisms (NSM and PSM), illustrating the correlation between the constant-torque magnitude and the relative angle θ_{pn}. Here, M_c denotes the combined torque generated by the reaction torques of the NSM (M_n) and PSM (M_p).

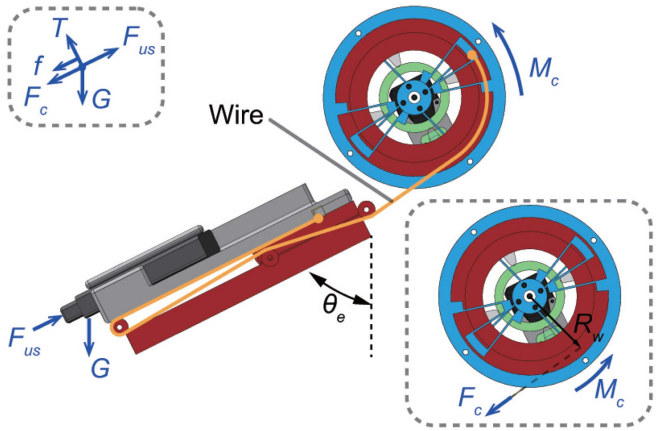


Fig. 4. Force analysis of the contact force exerted by the US probe, illustrating the transmission from the reaction torque to the output contact force.

where R_1 and R_2 satisfy the following relationships:

$$R_1 + R_2 = R + r \tag{10}$$

$$R_2 = R_1 + l_p. \tag{11}$$

In view of (7)–(9), f and m are derived as follows:

$$f = \frac{2m}{l_p} \tag{12}$$

$$m = \frac{6EI_pR_1R_2}{l_p^2(2R_2 + R_1)}\theta_p. \tag{13}$$

Considering the free-body diagram of the half cylinder, as shown in Fig. 2(b), the reaction torque of the PSM (M_p) with an angular deformation can be derived as

$$M_p = 2(fR_1 + m) = \frac{12EI_pR_1R_2(R_1 + R_2)}{l_p^3(2R_2 + R_1)}\theta_p. \tag{14}$$

Then, the stiffness of the PSM (K_{psm}) is given by

$$K_{\text{psm}} = \frac{M_p}{\theta_p} = \frac{12EI_pR_1R_2(R_1 + R_2)}{l_p^3(2R_2 + R_1)}. \tag{15}$$

Since the NSM and PSM are connect in parallel, the combined torque (M_c) is derived as

$$M_c = M_n + M_p. \tag{16}$$

Hence, by controlling θ_n to maintain the deformation within the constant-torque region (θ_r), the PM provides a constant-torque (M_{const}) to the winding reel, as shown in Fig. 3. In addition, the magnitude of M_{const} can be adjusted by tuning θ_{pn}. Based on the deformation direction required to enter the constant-torque region, there exists a corresponding adjusting direction for θ_{pn} that strengthens the deformation of the PSM. The allowable adjusting range of θ_{pn} can be expressed as follows:

$$[\theta_{\min}, \theta_{\max}] = [-\theta_{p,\max}, \theta_{p,\max} - \theta_{n,\max}] \tag{17}$$

where $\theta_{n,\max}$ and $\theta_{p,\max}$ denote the maximum angle of θ_n and θ_p due to the material limitation, respectively. In addition, $\theta_{n,\max}$ should be larger than or equal to $\theta_{p,\max}$.

Furthermore, the constant performance ($\Delta M\%$) is determined as the ratio of the torque variation and the average torque (M_{const}) within the constant-torque region, which is expressed as

$$\Delta M\% = \frac{\max M_c(\theta_r) - \min M_c(\theta_r)}{M_{\text{const}}} \times 100\%. \tag{18}$$

C. Modeling of Modular Torque-to-Force Part

As shown in Fig. 4, the force transmitted from the winding reel to the wire (F_c) is given by

$$F_c = \frac{M_c}{R_w}. \tag{19}$$

Referring to (14)–(19), the relationship of θ_{pn} and F_c can be expressed as

$$\theta_{\text{pn}} = \frac{R_w}{K_{\text{psm}}}(F_{c,0} - F_c) \tag{20}$$

where $F_{c,0}$ denotes constant contact force F_c obtained when $\theta_{\text{pn}} = 0$.

The US probe is subjected to multiple external forces, including the wire tension F_c, axial contact force F_{us}, supporting force from the slider T, equivalent gravitational force acting on the probe G, and equivalent frictional force f. Notably, the direction of f is governed by the instantaneous motion direction of the probe. On the basis of static force equilibrium, the expression for F_{us} is derived as follows:

$$F_{\text{us}} = F_c + G\cos\theta_e + f \tag{21}$$

where $G\cos\theta_e$ and f must be compensated in practical operation to eliminate gravitational interference and suppress force fluctuations induced by friction. Furthermore, the equivalent frictional force f mainly originates from the sliding friction of the slider and rotational friction of the pulleys, and its approximate expression is given by

$$f \approx f_e \sin\theta_e + f_c + M_h/(2R_w) \tag{22}$$

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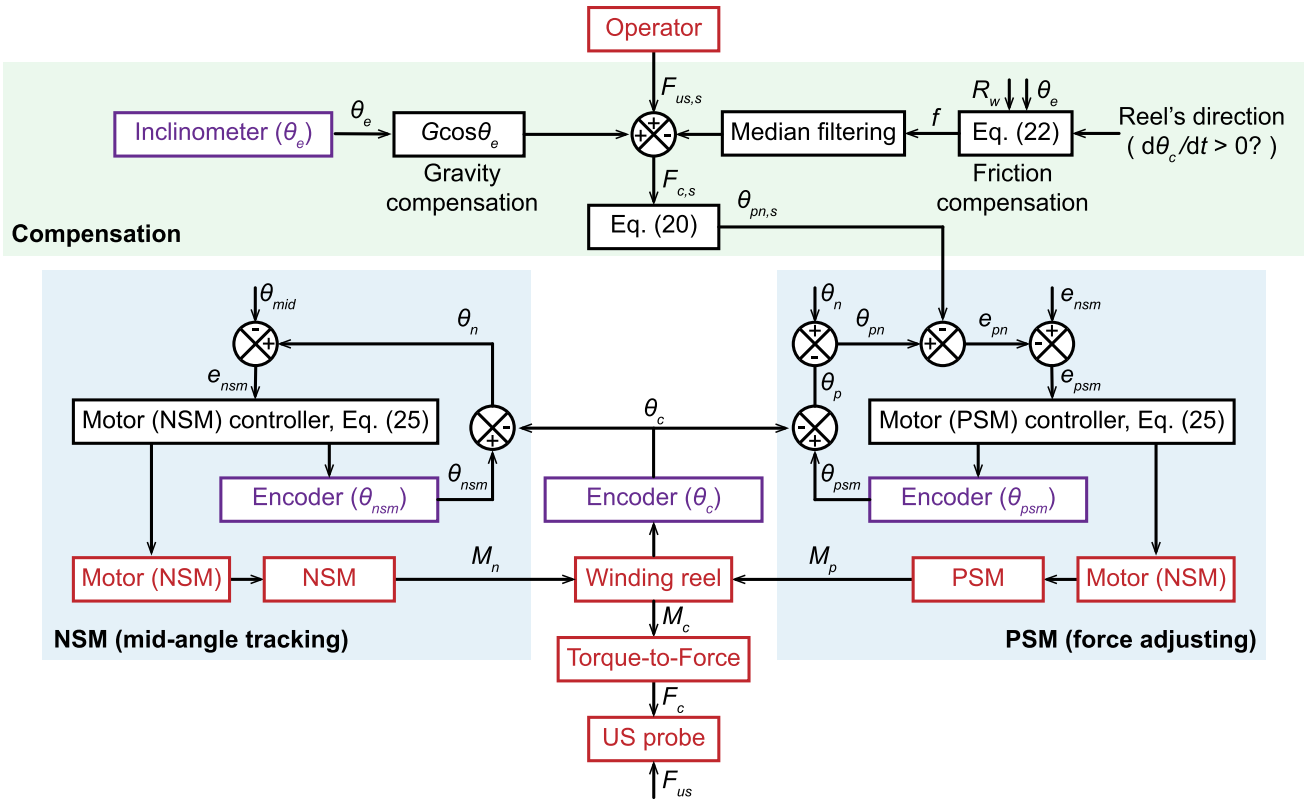


Fig. 5. Flowchart of the control system for the proposed RUSS, illustrating the signal transmission and processing pathways. The feedback signals, acquired from three encoders, reflect the deformation state of the PM. The control system inputs $F_{\mathrm{us},s}$ and θ_e are determined by the operator.

where f_e and f_c represent θ_e-dependent and θ_e-independent equivalent friction components, respectively, and M_h denotes the hysteresis torque deviation introduced by the PM.

D. Modeling of Active Deformation Control Part

Fig. 5 illustrates the flowchart of the control system of the proposed RUSS. The primary objective of the control system is to sustain the desired deformations of both NSM and PSM under diverse operating conditions and external disturbances, thereby enhancing the overall robustness of the RUSS. Accordingly, distinct US scanning tasks impose varying requirements in terms of the contact force and the tilt angle of the US probe. For this reason, the preset contact force ($F_{\mathrm{us},s}$) and inclination angle θ_e are manually configured by the operator within the developed system. Moreover, the measured direction of the frictional force f may shift frequently in practical operation, especially when the direction of low-frequency vibration reverses. To mitigate this issue, a median filter is adopted to smooth the friction compensation signal. Subsequently, the corresponded preset angle difference $\theta_{\mathrm{pn},s}$ is calculated based on (19) and (20).

According the feedback signals, the input signals of the motor controllers are determined as

$$e_{\mathrm{nsm}} = \theta_n - \theta_{\mathrm{mid}} \tag{23}$$

$$e_{\mathrm{psm}} = e_{\mathrm{nsm}} - e_{\mathrm{pn}} = e_{\mathrm{nsm}} - (\theta_{\mathrm{pn}} - \theta_{\mathrm{pn},s}) \tag{24}$$

where e_{nsm}, e_{psm}, and e_{pn} denote the real-time errors of the feedback angles to preset angles. The preset angular deformation of the NSM is chosen as the mid-angle (θ_{mid}) in the constant-torque region, as shown in Fig. 3. Then, the disturbance in both directions can be buffered by the PM.

In addition, a motor embedded controller (MEC) is integrated within the motor driver; however, this module is primarily engineered for point-to-point positioning operations. When implemented for continuous trajectory tracking, substantial positional lag occurs, since the actual position feedback consistently falls behind the reference trajectory. To rapidly eliminate motor positional deviations and suppress steady-state errors, an external PIC controller is employed to regulate the two motors driving the NSM and PSM separately. PIC represents a mature control scheme extensively deployed in motor position and speed regulation systems. Its stability is primarily determined by plant dynamics and the tuning of proportional and integral gains. It is notable that critical dynamic information of the MEC—including its internal control laws, sampling timing, and communication delays—are inaccessible closed-box parameters. Leveraging the strong adaptability and model-free nature of PIC, the proposed control strategy eliminates reliance on an accurate explicit system model and the internal control frequency of the MEC [40], [41]. Furthermore, the PIC control parameters are conservatively tuned based on system's dynamic boundaries to prevent oscillatory behavior and system divergence.

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The input control signals of the NSM (u_{nsm}) and PSM (u_{psm}) are given as

$$\begin{cases} u_{\text{nsm}} = K_p e_{\text{nsm}}(t) + K_i \int_0^t e_{\text{nsm}}(\tau) d\tau \\ u_{\text{psm}} = K_p e_{\text{psm}}(t) + K_i \int_0^t e_{\text{psm}}(\tau) d\tau \end{cases} \tag{25}$$

where K_p and K_i represent the proportional and integral coefficients of the PIC, respectively.

For comparison, we also investigated directly inputting the ideal values (θ_{mid} and $\theta_{\text{pn},s}$) to the motors of NSM and PSM, respectively. This comparative control scheme is denoted as the MEC.

III. Parameter Determination

This section elaborates on the determination of the PM's structural parameters via FEA and experimental evaluation. In accordance with the analytical relationships derived in Section II-B, a parameterized CAD model of the PM was established, as depicted in Fig. 1. For the convenience of rapid prototyping and experimental testing, the PM was fabricated via 3-D printing using acrylonitrile butadiene styrene (ABS) as the raw material. Most structural parameters were determined according to assembly constraints and the manufacturing capacity of the 3-D printer.

In this case study, R and r are set to 45.0 mm and 10.0 mm (see Fig. 2), respectively, with the overall dimension of the PM controlled within a reasonable range to avoid excessive sizing. As specified in Section II-B, the maximum angular deformation in the simulation $\theta_{n,\max}$ is defined as γ, which is strictly larger than the maximum angular deformation within the constant-torque regime. The variable parameters optimized for constant-torque performance are determined as α_n and l_p. Considering the 3-D printing manufacturing resolution, the minimum adjustment increments for these structural parameters are set to 0.1° and 0.1 mm, respectively.

Based on previous studies [37], [38], [39], [42], the constant-torque regime cannot be accurately predicted via FEA simulation alone, which is mainly attributed to prediction deviations of the buckling mode. Given the merits of 3-D printing in facile fabrication and rapid testing, an integrated tuning strategy combining FEA preliminary numerical screening and experimental evaluation is adopted to determine the variable parameters. The detailed tuning procedure is described as follows.

1) To achieve the longest constant-torque region under the given structural constraints, variable screening is performed via FEA simulation to determine the maximum feasible α_n. Besides, the minimum safety factor of the NSM under an angular deformation of γ is set to be larger than 1.1 to prevent material failure, and the length of the constant-torque region increases with rising α_n.
2) The NSM with the optimized maximum α_n is fabricated, its torque–deformation curve is measured experimentally, and the magnitude and effective range of the negative stiffness are calculated accordingly.
3) The value of l_p is derived via (15) to match the positive stiffness with the measured negative stiffness.
4) The PSM with the initially calculated l_p is fabricated and tested, and the value of l_p is sightly adjusted based on the experimental results.
5) For the tuned PSM, the maximum angular deformation $\theta_{p,\max}$ with a safety factor over 1.1 is determined via FEA simulation.

TABLE I
Structural Parameters and Mechanical Performances

Structural parameters			
Parameter	Value	Parameter	Value
R	45.0 mm	r	10.0 mm
α_n	17.8°	b_n	5.0 mm
h_n	1.0 mm	l_p	22.1 mm
b_p	5.0 mm	h_p	1.0 mm
Mechanical Performances			
Parameter	Value	Parameter	Value
M_{const}	218.16 N·mm	$\Delta M\%$	7.86%
Constant-torque region	[5.4°, 9.7°]	Allowable θ_{pn}	[–17.9°, 3.4°]
K_{psm}	7.71 N·mm/°	M_h	9.07 N·mm

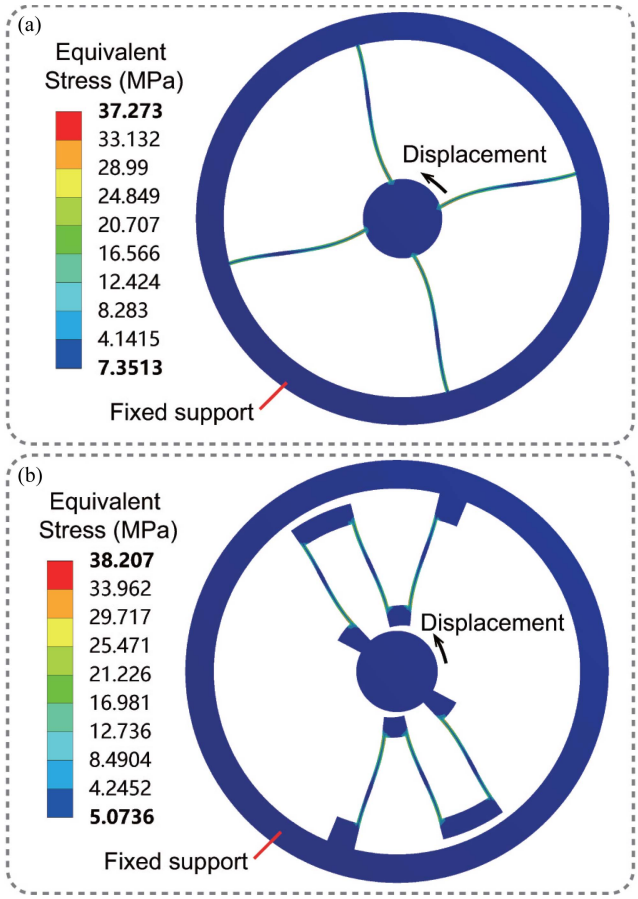


Fig. 6. Simulated stress distribution results of (a) NSM and (b) PSM at their maximum angular deformations.

Table I lists the optimal structural parameters and corresponding mechanical performances of the PM. Fig. 6 presents the simulated stress distribution results of the NSM and PSM under their maximum angular deformations. The maximum stresses in both components are lower than the yield stress of ABS (42 MPa), which fully meets the material safety requirements.

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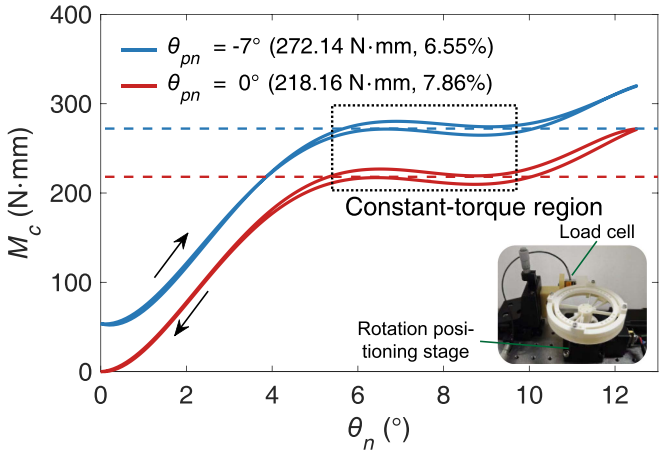


Fig. 7. Actual torque–deformation curves of the fabricated PM at various relative angles θ_{pn}, revealing the practical mechanical performance of PM. Dashed line: mean torque in the constant-torque region; brackets: (M_{const}, $\Delta M\%$); bottom-right inset: experimental setup.

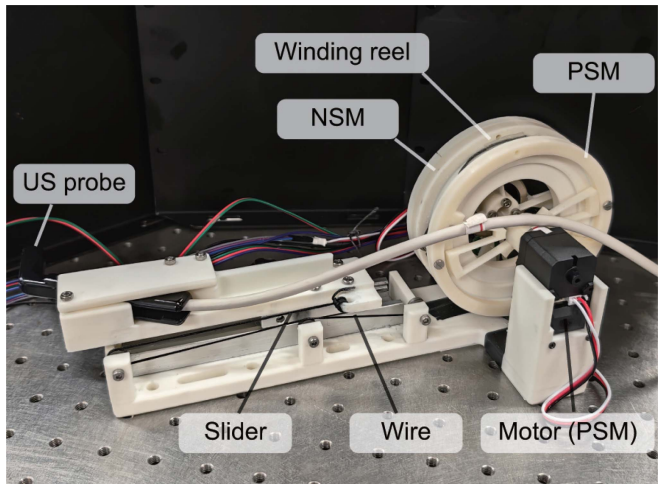


Fig. 8. Prototype photograph of the developed RUSS.

Based on the simulation results, the allowable adjustment range of θ_{pn} is calculated as [–17.9°, 3.4°].

According to the optimal structural parameters listed in Table I, a prototype of the PM was fabricated, as shown in the inset photo of Fig. 7. Furthermore, the torque measurement was carried out using a commercial force sensor (model: LSB200-25 lb, from FUTEK Advanced Sensor Technology, Inc.). The angular displacement of the PM was provided by a rotational positioning stage (model: Y200RA60, from Beijing Haijie Jiachuang Technology Company, Ltd), and a real-time controller (NI PXIe-8881, from National Instruments, Inc.) was adopted for data acquisition. In addition, Fig. 7 plots the measured torque–deformation curves of the PM under various values of θ_{pn}, which verifies that the constant-torque output can be tuned by regulating the angular difference between the NSM and PSM. Owing to the inherent material properties, a hysteresis effect exists between the loading and unloading curves, which constitutes the primary source of the torque fluctuation within the constant-torque regime. The average hysteresis difference between the upper and lower curves within the constant-torque region is calculated as 9.07 N·mm. Furthermore, the torque fluctuation rate $\Delta M\%$ for both loading and unloading processes is below 5% when θ_{pn} equals zero. Accordingly, half of the average hysteresis difference is introduced as a component of the equivalent friction (f) into the active control system [see (22)] to suppress force fluctuations. Besides, $\Delta M\%$ can be reduced to less than 5% by shifting the upper and lower hysteresis curves upward and downward, respectively. From Fig. 7, the adjustment coefficient (i.e., the stiffness of the PSM, K_{psm}) is experimentally determined as 7.71 N·mm/°. In addition, the stiffness K_{psm} obtained via FEA simulation is 7.67 N·mm/°, which is highly consistent with the experimental result. This favorable agreement arises because the FEA of the PSM eliminates the interference of the buckling effect. Consequently, FEA simulation proves to be reliable for predicting the mechanical performance of the PSM, and can be effectively employed to design a PSM that matches the NSM normally. The constant-torque regime ranges from 5.4° to 9.7°, with a valid rotation range of 4.3°.

IV. Experimental Testing and Results

This section presents the prototype fabrication and experimental setup of the proposed RUSS, followed by systematic experimental investigations to validate the effectiveness and overall performance of the proposed design scheme.

A. Prototype Fabrication

The winding reels and other connecting components of the system were fabricated via 3-D printing technology. System deformations and motions are regulated by servo motors (model: ST3215-HS) equipped with a dedicated motor driver (model: Bus Servo Adapter-A), both purchased from Shenzhen BoBo Electronics Company, Ltd. The angular values (θ_{nsm}, θ_{psm}) are acquired via the encoders embedded in the servo motors. Meanwhile, θ_c and θ_e are measured by a custom-designed magnetic encoder (model: AS5600) and a commercial inclinometer (model: GY-25) respectively, and all collected signals are processed by an Arduino Mega2560 control board. US scanning is implemented using a commercial portable US system (model: Terason u-smart 3300, Teratech), whose mainframe weighs only 6.7 kg and supports bedside and field diagnostic applications outside standard clinical settings. All components were integrated and assembled, as depicted in Fig. 8. The weight and overall dimensions of the end-effector are measured as 610 g and 287 $\times$ 122 $\times$ 129 mm^3, respectively, demonstrating excellent compactness and portability.

B. Evaluation of Controller Performance

To evaluate the performance of the proposed controller, a series of comparative experiments were conducted to validate the tracking performance regarding angular deformation, contact force adjustment, and gravity compensation. For the experimental tests, the RUSS was integrated with a winding reel of

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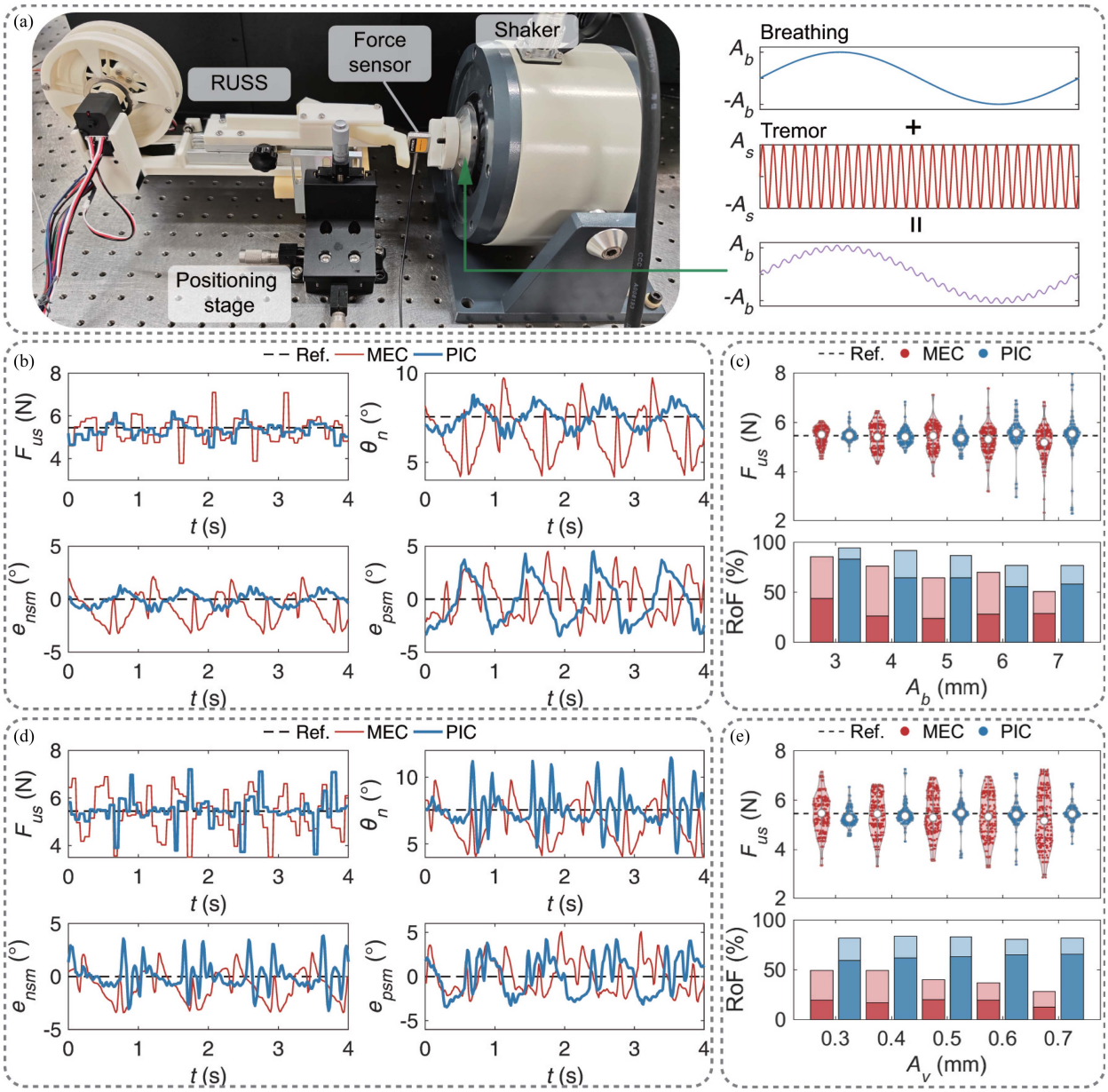


Fig. 9. Controller performance evaluation results. (a) Experimental setup. (b) Tracking results of the RUSS at $A_b = 5$ mm, demonstrating the controller performance under slow breathing-induced vibration. (c) Contact forces under various slow vibration amplitudes A_b without fast vibration, revealing the effect of A_b on contact force stability. (d) Tracking results of the RUSS at $A_b = 5$ mm and $A_v = 0.5$ mm, validating the controller performance under fast sudden tremor vibration. (e) Contact forces at $A_b = 5$ mm with various fast vibration amplitudes A_v, revealing the effect of A_v on contact force stability. Ref.: reference curve. RoF: Ratio of the contact force falling within the mean magnitude $\pm 5\%$ (dark-toned region) or $\pm 10\%$ (light-toned region).

radius $R_w = 40$ mm, and the corresponding experimental setup is illustrated in Fig. 9(a). The vibration disturbance of the contact surface is defined as $A_b \sin 2\pi v_b t + A_s \sin 2\pi v_s t$, where t refers to the experimental time; A_b and A_s represent the amplitudes of the low-frequency vibration (mimicking human breathing) and high-frequency vibration (simulating sudden body tremors), respectively, while v_b and v_s denote the corresponding vibration frequencies separately. Furthermore, the force discrepancies between the forward and backward probe movements were measured under various inclination angles θ_e, based on which the equivalent friction coefficients f_e and f_c are determined as 0.5 N and 0.09 N, respectively.

1) Demonstration of Angle Tracking: To verify the angular deformation tracking performance, the feedback signals and contact forces of the proposed RUSS under diverse vibrational disturbances were measured, as depicted in Fig. 9(a), with the inclination angle θ_e set to 0°. The control system parameters are listed in Table II, and these parameters were tuned via the trial-and-error method according to the experimental measurements. Correspondingly, the contact force results under various testing conditions are summarized in Table III.

Initially, the angle tracking performance of the proposed RUSS was tested under varying amplitudes of low-frequency vibration, with high-frequency vibration excluded in this set of

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TABLE II
COEFFICIENTS OF THE DEVELOPED CONTROL SYSTEM

Type	Parameter	Value
PIC	K_p	0.9
	K_i	3.0
Compensation	G	1.37 N
	f_e	0.5 N
	f_c	0.09 N
	M_h	9.07 N·mm

experiments. The frequency v_b was set to 0.5 Hz, which is higher than the typical human breathing frequency (12–20 breaths per minute). The angle and contact force tracking results under a vibration amplitude A_b of 5 mm are presented in Fig. 9(b). When utilizing the MEC, the angular error e_{nsm} is maintained within $\pm 3.30^\circ$, verifying that θ_n stays within the constant-torque regime throughout the testing process. However, the tracking performance of θ_p is comparatively poorer, with the angular error e_{psm} reaching $\pm 4.50^\circ$, which is attributed to the frequent fluctuations in friction compensation. Consequently, notable deviations between the measured contact force F_{us} and the reference value are observed.

In contrast, the experimental results obtained using the proposed dedicated PIC for the RUSS demonstrate superior angle tracking performance compared with the MEC. Specifically, the angular errors e_{nsm} and e_{psm} are restricted within $\pm 1.29^\circ$ and $\pm 4.51^\circ$, respectively. Meanwhile, nearly all peak deviations of F_{us} appear at the moments when e_{psm} reaches its peak values. Furthermore, the root-mean-square value of e_{nsm} (0.59°) is considerably lower than that of the MEC (1.67°), as illustrated in Fig. 9(b). In addition, as the vibration amplitude A_b increases from 3 to 7 mm, the proposed controller increases the ratio of contact force (RoF) within $\pm 5\%$ and $\pm 10\%$ of the mean value by average margins of 34.98% and 15.94%, respectively (see Table III). As depicted in Fig. 9(c), the RoF exhibits a declining trend with the rise of A_b, indicating that the force tracking performance of the MEC gradually degrades under large-amplitude vibrations. This phenomenon stems from the limited rotational speed (106 r/min) of the adopted servo motor.

Second, the tracking performance of the proposed RUSS was tested under varying amplitudes of high-frequency vibration, with the results presented in Fig. 9(d) and (e). In this group of experiments, the low-frequency vibration amplitude A_b was set to 5 mm, with a corresponding frequency v_b of 0.5 Hz. The peak-to-peak vibration displacement reaches 10 mm, which exceeds the chest wall displacement during normal breathing [43]. Meanwhile, the high-frequency vibration frequency v_s was configured as 15 Hz, larger than the frequency range of common physiological tremors (4–12 Hz) [44]. As illustrated in Fig. 9(d), the contact force fluctuations intensify at the moments when the direction of low-frequency vibration reverses, signifying that the angle tracking performance of the PIC declines slightly under the superposition of high-frequency tremors. Nevertheless, as depicted in Fig. 9(e), the RoF values within $\pm 5\%$ and $\pm 10\%$ of the reference contact force both decrease by less than 5% when using the proposed PIC (see Table III), which verifies

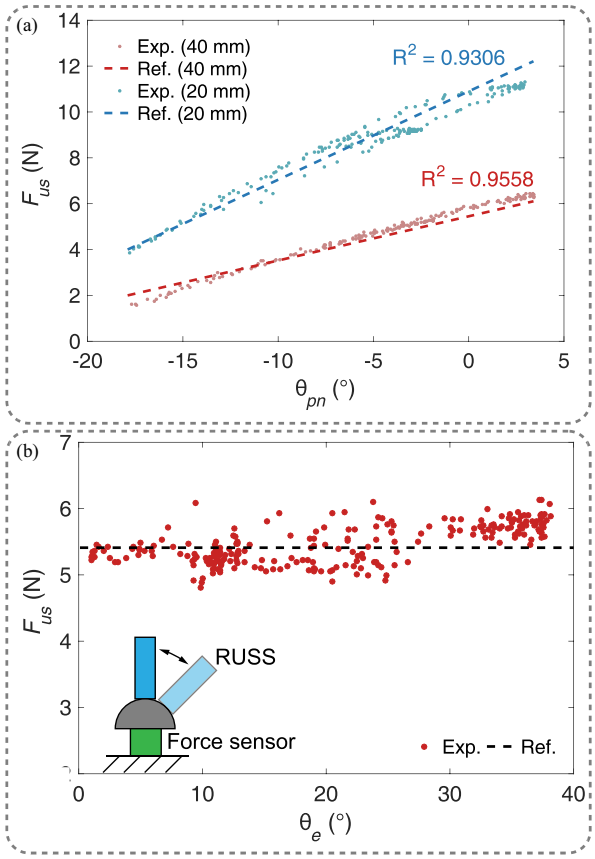


Fig. 10. Force adjusting and gravity compensation test results. (a) Tracking results of constant-force magnitude regulation, demonstrating the force adjustment function of the proposed system. (b) Experimental results of automatic gravity compensation, revealing the RUSS performance under varying operating angles.

that the designed PM can effectively suppress force fluctuation induced by sudden physiological tremors. By comparison, the MEC exhibits far inferior tracking and force stabilization performance relative to the PIC. This outcome arises because the system spends a longer duration outside the constant-force region when controlled by the MEC, preventing the PM from buffering the high-frequency tremor disturbances effectively. This further validates the superiority of the proposed dedicated controller. Moreover, as listed in Table III, the RoF values within the maximum clinically acceptable force threshold of $\pm 20\%$ [15] exceed 85% even under the combined disturbance of breathing and continuous physiological tremors, proving the strong clinical application potential of the proposed RUSS.

2) Demonstration of Force Magnitude Adjustment: To validate the feasibility of regulating the constant contact force magnitude via tuning θ_{pn}, the active control system was employed to track a 0.1 Hz sinusoidal reference force trajectory, as depicted in Fig. 10(a). When θ_{pn} is adjusted from -17.9° to 3.4°, the contact force magnitude varies from 1.6 to 6.4 N with a winding reel radius R_w of 40 mm, and this variation trend aligns well with the theoretical model given in (20). The minor deviations between the experimental data and the reference trajectory are mainly ascribed to assembly tolerances. The experimental results effectively verify the feasibility of

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TABLE III
SUMMARY OF CONTROLLER PERFORMANCE EVALUATION RESULTS

	A_b		3 mm	4 mm	5 mm	6 mm	7 mm
Fig. 9(c)	Mean ± SD	MEC	5.41 ± 0.38 N	5.42 ± 0.51 N	5.37 ± 0.62 N	5.23 ± 0.70 N	5.05 ± 1.00 N
		PIC	5.49 ± 0.25 N	5.44 ± 0.36 N	5.32 ± 0.34 N	5.48 ± 0.76 N	5.45 ± 0.96 N
	RoF of ±5%	MEC	43.91%	26.20%	24.00%	28.04%	28.78%
		PIC	83.39%	64.58%	64.58%	55.35%	57.93%
	RoF of ±10%	MEC	85.61%	76.38%	64.58%	69.74%	50.92%
		PIC	94.46%	91.88%	86.72%	77.12%	76.75%
	RoF of ±20%	MEC	100.00%	98.16%	93.00%	87.45%	83.03%
		PIC	100.00%	97.05%	100.00%	88.19%	87.45%
	A_v		0.3 mm	0.4 mm	0.5 mm	0.6 mm	0.7 mm
Fig. 9(e)	Mean ± SD	MEC	5.42 ± 0.80 N	5.37 ± 0.82 N	5.36 ± 0.89 N	5.28 ± 0.93 N	5.21 ± 1.13 N
		PIC	5.31 ± 0.40 N	5.45 ± 0.51 N	5.50 ± 0.63 N	5.44 ± 0.68 N	5.45 ± 0.44 N
	RoF of ±5%	MEC	19.56%	16.97%	19.93%	19.56%	12.55%
		PIC	59.41%	61.99%	63.10%	64.94%	65.31%
	RoF of ±10%	MEC	49.45%	49.08%	39.85%	36.90%	28.41%
		PIC	81.92%	83.76%	83.05%	80.44%	81.92%
	RoF of ±20%	MEC	86.72%	81.18%	71.22%	74.17%	56.09%
		PIC	97.78%	92.99%	88.19%	85.24%	93.36%

tuning the contact force magnitude adjustment by adjusting θ_{pn}. In addition, no material failure occurs in the PM within the allowable range of θ_{pn}, which further confirms the validity of the FEA simulation results presented in Section III.

To demonstrate the modular design for switching the constant-force range via replacing the winding reel, the corresponding constant-force ranges were measured with a winding reel radius R_w of 20 mm, as plotted in Fig. 10(a). Under this parameter, the contact force magnitude ranges from 3.8 to 11.3 N. The experimental results reveal that reducing the winding reel radius contributes to an increase in the average constant-force magnitude, which is consistent with the theoretical derivation of (19). Accordingly, the proposed RUSS can satisfy diverse contact force demands for various US scanning tasks via straightforward replacement of the modular winding reel. Furthermore, the simplified modular replacement procedure is also user-friendly for inexperienced operators.

3) Demonstration of Gravity Compensation: To evaluate the gravitational effect under varying scanning inclinations, the equivalent gravity of the end-effector components was measured as 1.37 N. Based on the real-time measured inclination angle θ_e, the gravitational disturbance is automatically compensated by tuning the angular difference θ_{pn}.

To verify the effectiveness of the proposed gravity compensation scheme, experiments were performed to measure the contact force F_{us} under manual inclination adjustment of the end-effector, as illustrated in Fig. 10(c). The experimental results indicate that the contact force magnitude remains stable under varying inclined angles after implementing gravity compensation, which validates the reliability of the RSSS for US scanning tasks across a wide operating angle range. Furthermore, this result also demonstrates the system's capability for real-time adjustment of the contact force magnitude, as varying the inclined angle inherently changes the required force provided from the PM.

In the absence of friction and gravity compensation, the constant-force range is calculated as [2.00 N, 6.11 N] with a winding reel radius R_w of 40 mm. After incorporating the compensation terms, the constant-force range is narrowed due to the friction compensation, and its lower bound is further elevated by the gravity compensation. Consequently, the revised constant-force range with dual compensation is determined as [4.07 N, 5.41 N]. The predicted constant-force ranges corresponding to various R_w are summarized in Table IV. It can be observed that gravity compensation imposes a more significant impact on the constant-force range when using a small R_w (i.e., low-level constant-force output), which is attributed to the fact that the gravity compensation magnitude accounts for a large proportion of the total constant-force output under a small reel radius. In addition, the maximum working displacement is defined as the smaller value between the perimeter of the assembled winding reel and the travel length of the sliding guide rail. For the experimental setup adopted in this work, the travel length of the sliding guide rail is set to 90 mm.

TABLE IV
PREDICTED CONSTANT-FORCE RANGES AT VARIOUS WINDING REEL RADII R_w

R_w	Constant-force range
10 mm	[10.43 N, 23.39 N]
20 mm	[6.19 N, 11.40 N]
30 mm	[4.78 N, 7.40 N]
40 mm	[4.07 N, 5.41 N]
50 mm	[3.65 N, 4.21 N]

C. Tests Results of RUSS for US Scanning

To demonstrate the vibration isolation performance and universality of the proposed RUSS, experimental studies were conducted to acquire US images using a custom-built silicone phantom. The model, made of silicone rubber materials (Ecoflex 00-10), contained three embedded channels and was chosen owing to its low Young's modulus (55 kPa). Such material compliance renders the phantom highly deformable, which helps distinctly reveal the impact of contact force variation on US

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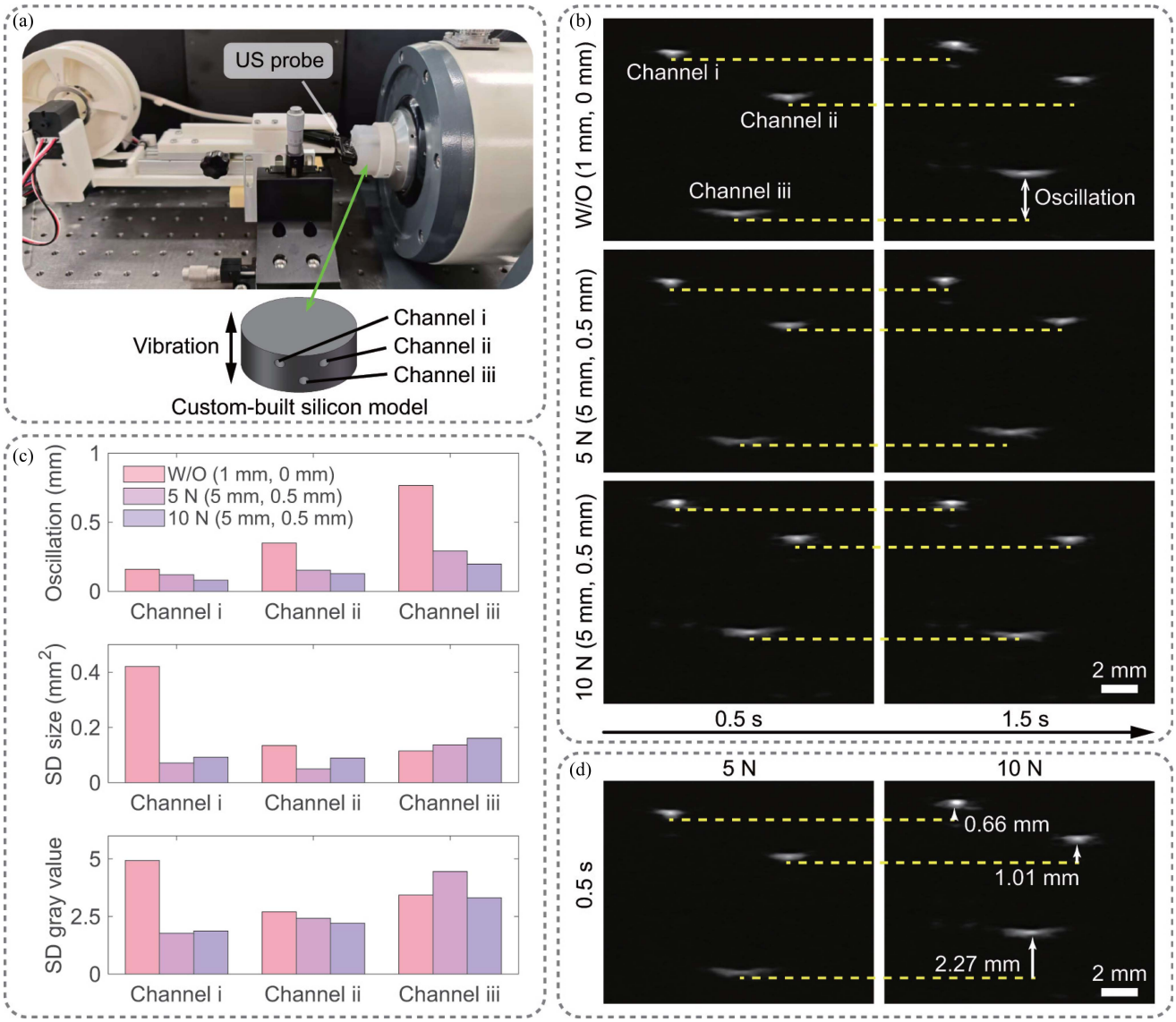


Fig. 11. US imaging test results. (a) Experimental setup. (b) US images during the scanning with/without the RUSS. (c) Mean oscillations, size changes and gray value changes of the measured channels with/without the RUSS, demonstrating the vibration isolation performance. (d) US images with various contact forces, illustrating the influence of force magnitude on the US imaging. Dashed line: reference position. Parentheses: (A_b, A_s).

imaging quality under different testing conditions. In addition, its Young's modulus is close to that of human gluteal subcutaneous and muscle tissues (62.2 kPa) under low load (5 N) [45], enabling high-fidelity simulation of clinical ultrasonic scanning scenarios.

1) US Imaging With Vibration Isolation: To evaluate the vibration isolation performance of the proposed RUSS, the centroid positions of the embedded channels in US images acquired with and without the RUSS were measured via computer image processing, as depicted in Fig. 11(a). When equipped with the RUSS, the contact surface vibration is set as $5\sin 1.88t + 0.5\sin 94.2t$ [mm]. For the control group without the RUSS, the contact surface vibration is fixed at $1\sin 1.88$ [mm], and the US probe remains stationary and directly attached to the silicon phantom throughout the entire scanning process. In addition, the depth of channel i was set as the focus depth of the US scanning.

Fig. 11(b) (see Movie 1 in the Supplementary Material) displays the periodic distortion of the acquired US images. Without the assistance of the RUSS, the three embedded channels present severe positional oscillations, which are further aggravated with the increase of scanning depth due to the elevated tissue compression volume. In contrast, the proposed RUSS remarkably suppresses such positional deviations by maintaining stable contact forces at 5 N and 10 N, even under more intensive contact surface vibrations. To quantify the vibration isolation efficacy, channel displacement oscillations were quantitatively measured via image processing algorithms. The average oscillation amplitudes of the three channels under different testing conditions were shown in Fig. 11(c), and the quantitative results

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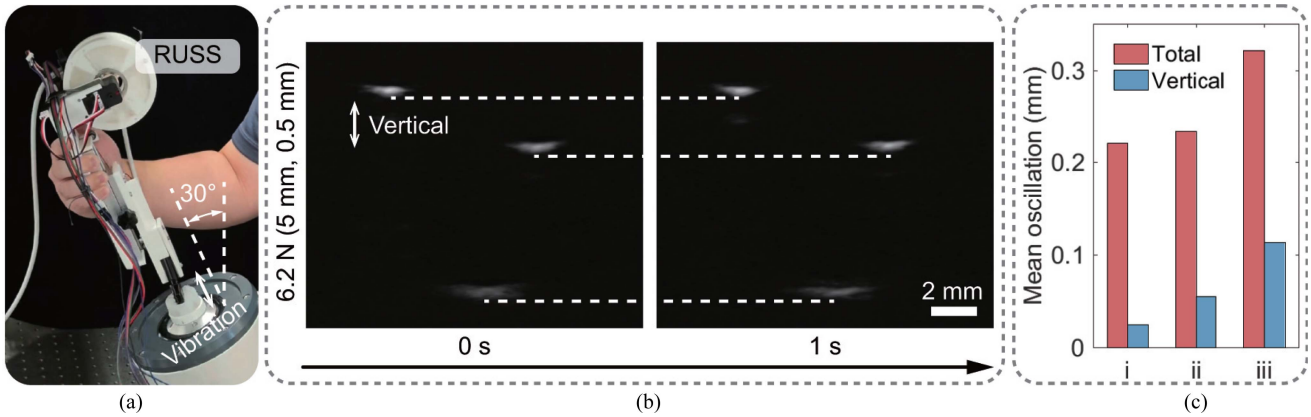


Fig. 12. Handheld US imaging test results. (a) Experimental snapshot. (b) US images during scanning. Dashed line: reference position. Parentheses: (A_b, A_s). (c) Channel oscillations, demonstrating the system performance under handheld operation.

validate that the proposed RUSS effectively attenuates the positional oscillations via its superior vibration isolation performance. Furthermore, the vibration suppression effect becomes increasingly prominent as the scanning depth of the target structure increases. Furthermore, to quantify the influences of the image blurring or edge artifacts from the residual oscillations, the standard derivations (SDs) of the measurement sizes and gray values are illustrated in Fig. 11(c). It is observed that both the variations of measurement size and gray value for channel i have been significantly reduced via the RUSS. However, the size variation for channel iii is increased. This reveals a limitation of the current system that the residual oscillations can be more detrimental to certain image quality metrics than low-frequency drift, especially for the deep tissue.

For severe positional oscillations during US scanning, it will cause the target anatomical structure to periodically shift out of the imaging plane, which can further lead to image discontinuity, inaccurate lesion diameter measurements, and interrupted Doppler spectrum acquisition. By contrast, minor residual oscillations with high frequency, though less disruptive to the view of core anatomical structures, will still induce image blurring and edge artifacts, particularly in deep tissue regions. Therefore, striking a balance between compensating these two conditions is crucial and requires further investigation.

Fig. 11(d) illustrates the effect of contact force magnitude on the US image quality. The results reveal that the compression degree of the embedded channels rises with the increase of contact force, while a relatively larger contact force also enhances the imaging sharpness of deep-seated targets (i.e., Channel iii). This phenomenon further verifies the necessity of maintaining a stable and appropriate contact force during clinical US scanning.

2) Demonstration of Handheld Operation of RUSS: To validate the portability and handheld operability of the proposed RUSS, an inexperienced operator was recruited to perform US scanning on the silicone phantom using the RUSS via handheld operation, as illustrated in Fig. 12(a). The handheld capability of the system drastically reduces the deployment complexity, for instance, eliminating the demands for base fixation and robotic arm calibration. The inclination angle of the silicon phantom is 30°, the contact surface vibration is configured as $5\sin 1.88t + 0.5\sin 94.2t$ [mm], and the preset contact force magnitude is 6.2 N.

Fig. 12(b) (see Movie 2 in the Supplementary Material) presents the US images acquired during the scanning under handheld operation. The vertical oscillation of the target structure is proven to be minor, which is even much smaller than the corresponding results presented in Fig. 11. This improvement can be attributed to the manual compensation of the operator who consciously maintains stable contact pressure during scanning. Nevertheless, apparent horizontal shifting of the imaging target occurs due to the slight movement of the operator's hand. As depicted in Fig. 12(c), the overall structural oscillation amplitude remains close to the results shown in Fig. 11. Furthermore, considering that vertical oscillation poses a dominant impact on the imaging clarity of scanned targets, the experimental results confirm that inexperienced operators can conduct stable US scanning smoothly using the proposed handheld RUSS. Benefiting from the lightweight end-effector, the proposed RUSS can support US scanning tasks in various working scenarios, particularly bedside and outdoor clinical diagnoses. In addition, it remarkably reduces the professional skill threshold required for high-quality US scanning.

3) Demonstration of Teleoperation of RUSS: Apart from standalone handheld operation, the proposed RUSS can be readily integrated with a robotic arm to realize stable force control during teleoperated remote diagnosis. For experimental validation, the RUSS prototype was mounted on a robotic arm (UR5, from Universal Robots A/S), and the end-effector motion was governed via a haptic interface (model: Geomagic Touch, from 3-D Systems Corp.). The connection architecture of the teleoperated RUSS system is illustrated in Fig. 13(a). A position-position mapping strategy was adopted for the teleoperation test. The operator manipulated the haptic interface to move the end-effector to the target scanning surface, after which the RUSS was activated to automatically maintain stable contact force between the US probe and the sample surface throughout the scanning procedure.

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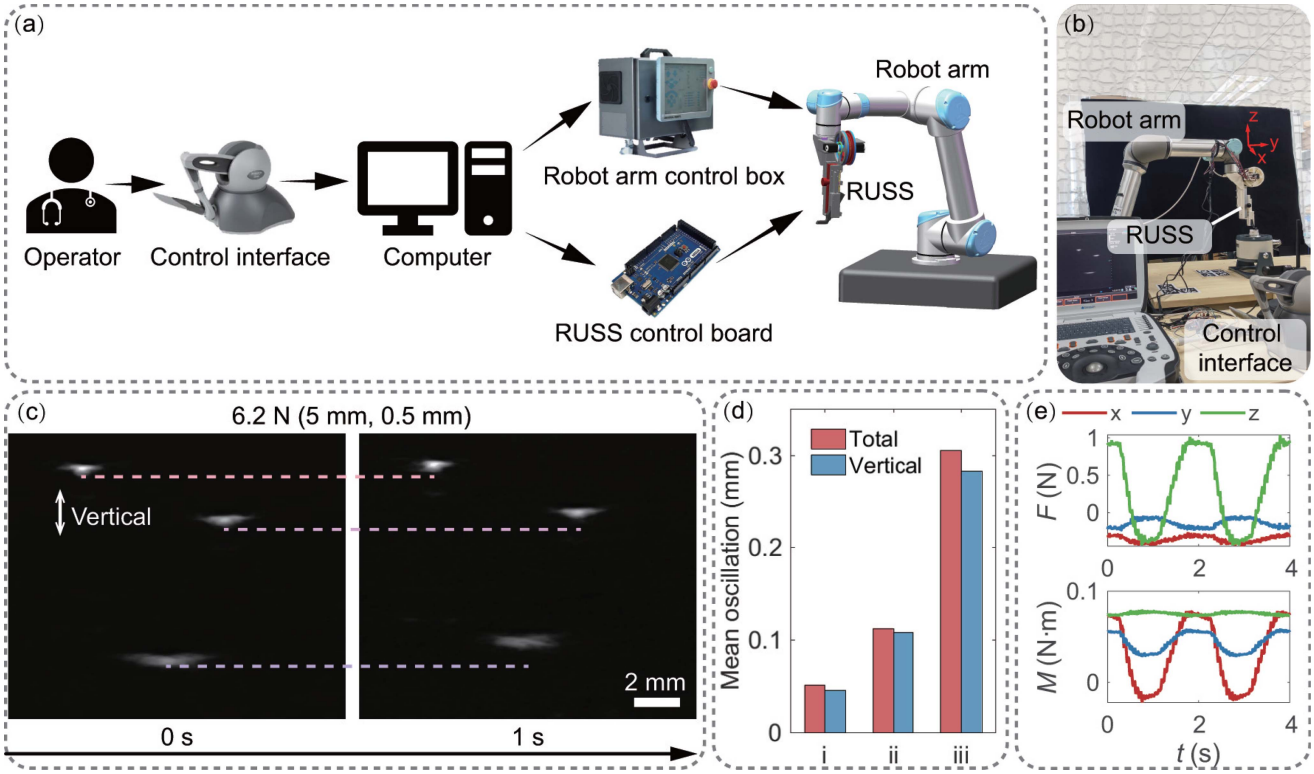


Fig. 13. Teleoperated US imaging test results. (a) Connection scheme of the teleleoperation RUSS. (b) Experimental snapshot. (c) US images during scanning. Dashed line: Reference position. Parentheses: (A_b, A_s). (d) Channel oscillations, revealing the system performance under teleoperation. (e) Reaction forces/torques at the end of the robotic arm.

The inexperienced operator performed remote US scanning on the same vibrated silicone phantom using the teleoperated RUSS system. The vibrational parameters and preset contact force were set identical to those in the handheld test, with the scanning surface adjusted to a horizontal orientation. The experimental setup is shown in Fig. 13(b), and the acquired US images are depicted in Fig. 13(c) (see Movie 2 in the Supplementary Material). The overall oscillation amplitude of the embedded channels remains comparable to the results presented in Fig. 11, verifying the feasibility of the proposed RUSS for reliable remote diagnostic scanning. Furthermore, in contrast to handheld operation, no horizontal shifting of the target structure is observed owing to the position-locking function of the robotic arm, and the total oscillation amplitude is nearly consistent with the vertical oscillation component, as plotted in Fig. 13(d). In addition, the robotic arm features outstanding motion repeatability. Accordingly, high long-term scanning repeatability can be achieved by recording the trajectory generated during the initial scan and reproducing it in subsequent scanning sessions. This functionality is highly beneficial for monitoring the long-term progression of potential lesions and realizing longitudinal diagnostic evaluation.

Moreover, we measured both reaction forces and torques at the end of the robot arm during the scanning, as shown in Fig. 13(e). It is observed that there are periodic variations in the reaction forces/torques during scanning. Specifically, the average force along the z-axis is 0.31 N, which is close to the predicted value of 0.22 N derived from the controlled contact force and the weight. Notably, owing to the offset contact point, there is a relatively large variation in the torque around the x-axis (0.10 N·m). Since such magnitude is much smaller than the maximum acceptable torques for male workers [46], the torque has slight effect to the operator. Furthermore, the remaining reaction forces and torques exhibited small magnitudes and small variations (less than 0.03 N·m), which may result from the alignment errors and the asymmetric design. This further verifies the viability of the proposed RUSS for handheld operation, since the operator experiences no substantial additional loads other than the contact force.

D. Practical Performance on Scanning Tissue

To assess the repeatability of US images acquired by the proposed RUSS in a practical setting, repeated scanning tests were performed on the cephalic vein of a healthy volunteer. An inexperienced operator conducted all scanning tasks via handheld operation, as shown in Fig. 14(a). The RUSS was removed and repositioned between consecutive scanning trials, and the volunteer remained in a steady resting state throughout the tests, with the contact force stably maintained at 6.2 N. The acquired US images are displayed in Fig. 14(b) (see Movie 3 in the Supplementary Material). The scanned

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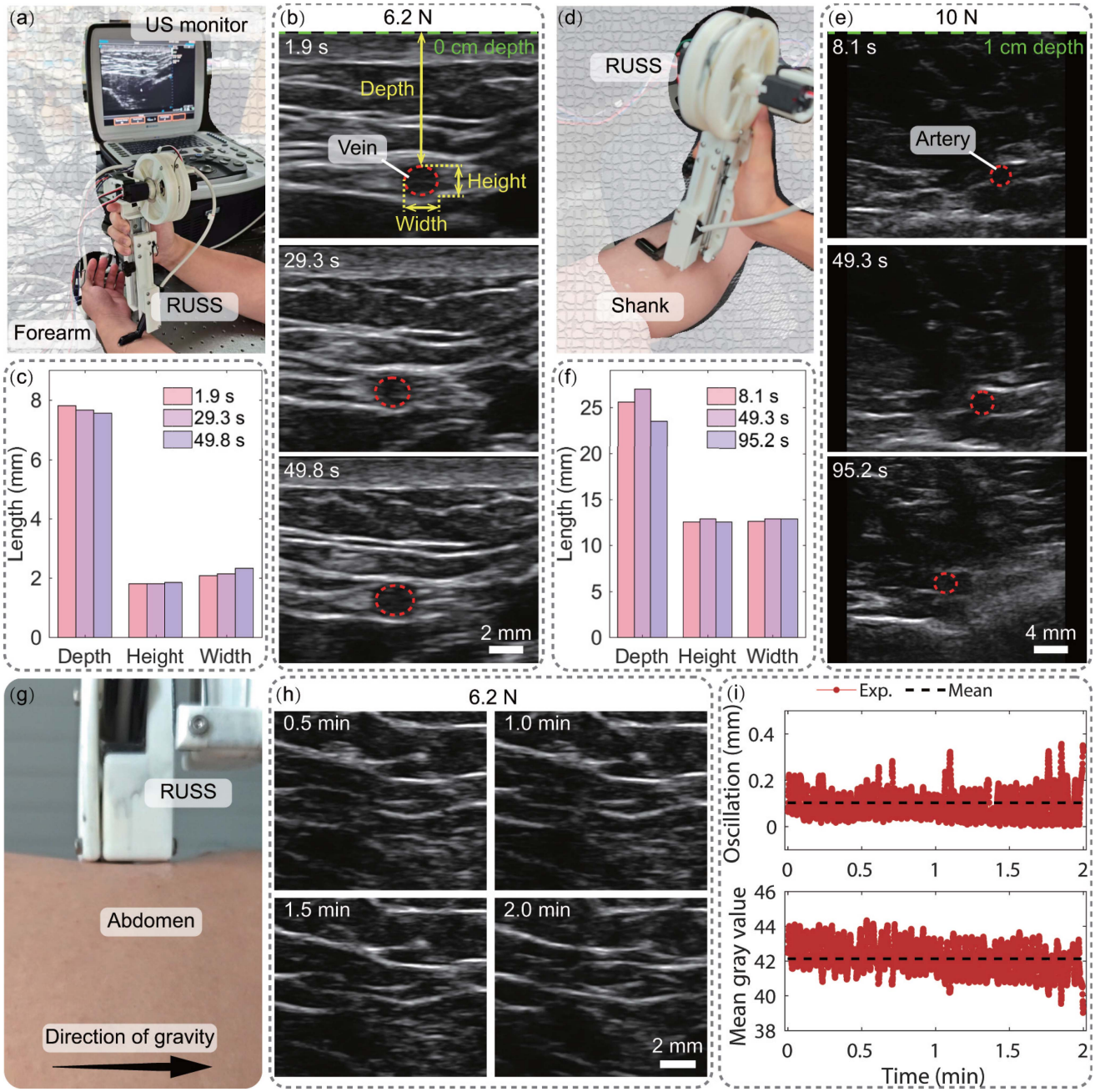


Fig. 14. Tissue scanning test results. (a) Experimental snapshot of forearm scanning. (b) US images of target vein. (c) Veins' depth and shape measurements, demonstrating the US imaging repeatability. (d) Experimental snapshot of shank scanning. (e) US images of the target artery. (f) Arteries' depth and shape measurements, indicating the US imaging repeatability. (g) Experimental snapshot of abdominal scanning under deep breathing. (h) US images of the abdomen. (i) Centroid oscillation and mean gray value histories, revealing the RUSS performance with moving scanned surface. Black dashed line: mean of experimental data.

venous structures exhibit high consistency in terms of both depth and profile across repeated trials. To quantify the scanning repeatability, the depth, height, and width of the targeted vein [as marked in Fig. 14(b)] were measured quantitatively, with the statistical results presented in Fig. 14(c). The measured parameters only show negligible discrepancies, with corresponding standard deviations of 0.13 mm, 0.03 mm, and 0.13 mm, respectively. These quantitative results confirm that the proposed RUSS effectively improves the repeatability of US scanning.

To further verify the system's universal applicability across different anatomical regions, additional repeated scanning tests were conducted on the anterior tibial artery of the same volunteer in a resting state, as depicted in Fig. 14(d). The contact force was adjusted to 10 N, considering the deeper anatomical position of the anterior tibial artery compared with the cephalic vein. The corresponding US images are presented in Fig. 14(e) (see Movie 3 in the Supplementary Material). Slightly larger intertrial discrepancies are observed in the acquired images, owing to the greater surface curvature of the shank relative to the forearm

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and minor inconsistencies in scanning angle. Despite this, the depth and profile of the targeted artery remain highly consistent across all trials, as shown in Fig. 14(f). These results validate the broad applicability of the proposed RUSS for diverse anatomical scanning scenarios, benefiting from its adjustable contact force feature.

To validate the system's practical performance in acquiring stable US images under intensive vibrational disturbances, in vivo scanning was performed on the volunteer's abdomen during deep breathing in a sitting posture, as illustrated in Fig. 14(g). This testing scenario simulates clinical conditions where patients with limited mobility cannot maintain a stable posture or fully cooperate with operators. The acquired US sequences are displayed in Fig. 14(h) (see Movie 3 in the Supplementary Material). The depth and gray-scale values of the anatomical structures remain remarkably stable throughout the scanning process. For quantitative evaluation, the centroid oscillations and mean gray-scale values of all imaging frames were measured, with the results plotted in Fig. 14(i). The quantitative data reveal only minor fluctuations in imaging quality under continuous deep breathing. Specifically, the mean centroid oscillation of the acquired images is 0.10 mm, with a standard deviation of 0.94 for the mean gray-scale value. This outcome firmly verifies the capability of the proposed RUSS in maintaining stable imaging quality under continuous dynamic disturbances of the scanned tissue surface.

E. Further Discussion

Compared with the previous work reported in [37], the proposed RUSS exhibits slightly weaker force control performance, as reflected by the minor force peaks in Fig. 9(c) and (e). This limitation is mainly attributed to the frictional effects introduced by the transmission components. Nevertheless, these components are indispensable for realizing the modular design and compact, slim scanning end-effector, which streamline the constant-force range switching procedure and improve patient comfort during clinical examinations. In addition, adopting a rotational constant-torque mechanism (instead of a conventional linear constant-force configuration) drastically reduces the overall weight and dimensions of the system by removing the bulky ball screw assembly. This structural optimization enables true handheld manipulation and greatly enhances the system portability for diversified clinical scenarios. Moreover, experimental results on the soft silicone phantom reveal that the impact of environmental vibrations on US imaging quality becomes more significant with decreased tissue stiffness. Under the same magnitude of contact force fluctuation, the positional oscillation of US imaging features is considerably more severe in softer phantoms, indicating that efficient vibration isolation is particularly critical for scanning patients with high body mass index. This finding may also partially explain the commonly observed poor US imaging quality in obese patients [47], [48].

Furthermore, the performance of the adopted servo motor imposes a critical impact on the overall control performance of the RUSS. The maximum rotational speed of the motor determines the allowable scanning traversal speed; an excessively high traversal rate will result in the loss of stable contact between the US probe and tissue surface. Therefore, utilizing servo motors with higher rotational speed is beneficial for enhancing force control robustness. However, motor selection must comprehensively balance multiple performance metrics, including output torque, weight, and physical volume, rather than only focusing on rotational speed. Ignoring these constraints will render the system inoperable or unfeasible for handheld operation due to excessive weight or bulky dimensions.

US probes of various sizes and weights are applied for different diagnostic applications. For heavier US probes, the corresponding gravity compensation magnitude G in the control system must be increased accordingly. Notably, increasing the probe weight will narrow the effective constant-force range for all modular winding reel sizes. For instance, when equipped with a 300 g US probe, the available constant-force ranges are reduced to [13.08 N, 23.39 N] and [8.84 N, 11.40 N] for winding reels with radii of 10 mm and 20 mm, respectively.

Nevertheless, the system has certain inherent limitations. Residual oscillations cause severe blurring in deep tissue areas and degrade observation quality. Accordingly, it remains debatable whether acceptable large-amplitude oscillations are diagnostically superior to strict vibration isolation for these specific regions. In addition, it is noteworthy that when the probe axis is tilted relative to the surface normal of the patient's body (e.g., during cardiac surgery), the axial force exerted by the RUSS decomposes into normal and tangential components. This impedes independent normal-force control of the probe, thereby necessitating inclination angle measurement for accurate force decomposition.

The proposed RUSS holds great promise for maintaining stable contact force in remote diagnostic scenarios, thereby improving global medical accessibility. By eliminating the need for on-site presence of specialized sonographers, high-quality US scanning can be delivered to patients in remote and underserved communities via the teleoperated RUSS integrated with a robotic arm. In addition, owing to its superior portability, the RUSS can be manually operated by frontline responders in field first-aid scenarios. Under remote expert guidance, nonspecialized operators can adjust the system constant-force to perform rapid bedside US scanning, enabling timely detection of life-threatening conditions such as internal hemorrhage and buying precious time for subsequent emergency treatment. Approval of all ethical and experimental procedures and protocols was granted by the Research Ethics Committee of the University of Macau under application no. APP-ARE-057.

V. Conclusion

This article presents the mechanism design, control framework, and experimental validation of a novel general-purpose robotic-assisted US scanning system. Featuring a lightweight end-effector, hybrid force control strategy, and modular force-range switching, the proposed RUSS possesses outstanding vibration isolation performance and a wide adjustable constant-force range, enabling stable operation across diverse US scanning scenarios. Its compact, lightweight

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design supports both handheld and teleoperated manipulation, expanding US diagnosis to bedside, remote and field clinical settings. The mechanical structure and closed-loop control system of the RUSS are systematically designed and analyzed, with a functional prototype fabricated for comprehensive experimental testing. Experimental results consistently verify that the proposed RUSS achieves stable contact force control, high imaging repeatability, and strong anti-interference capability, making it a reliable solution for various clinical US scanning tasks. In future work, extensive in vivo scanning tests will be conducted under professional guidance to further validate the system's clinical adaptability. Future research will also focus on automating the constant-force range switching module to reduce manual operation and improve the system intelligence.

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